Linear programming for T-avoiding spherical codes

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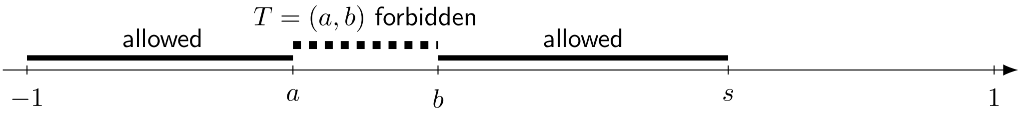
Motivation

Classical question

How large can a spherical code be when its pairwise inner products are bounded above by s?

Our setup: forbidden inner products

We additionally prescribe a set $T \subset [-1, s)$ which pairwise inner products must avoid.



Basic objects

Definition (Spherical s-code)

A finite set $C \subset \mathbb{S}^{n-1}$ is a spherical (n, N, s)-code if

$$|C| = N, \qquad s = s(C) := \max\{\langle x, y\rangle : x, y \in C,\ x \neq y\}.$$

Definition (Inner product set)

$$I(C) := \{\langle x, y\rangle : x, y \in C,\ x \neq y\}.$$

Definition (T-avoidance)

For $T \subset [-1, s)$, an s-code C is T-avoiding if

$$I(C) \cap T = \emptyset.$$

Spherical designs

Spherical τ-design

A code $C \subset \mathbb{S}^{n-1}$ is a τ-design if

$$\int_{\mathbb{S}^{n-1}} p(x)\, d\sigma_n(x) = \frac{1}{|C|} \sum_{x \in C} p(x)$$

for every polynomial p of total degree at most τ.

T-avoiding designs

A spherical design is T-avoiding if

$$I(C) \cap T = \emptyset.$$

Energy and universal optimality

h-energy

For a potential $h : [-1, 1] \to (-\infty, +\infty]$,

$$\mathcal{E}^h(C) := \sum_{x \neq y \in C} h(\langle x, y \rangle).$$

T-avoiding universal optimality

A T-avoiding code is universally optimal if it minimizes $\mathcal{E}^h$ among all T-avoiding codes of the same cardinality for every absolutely monotone potential h.

Positive definite functions on the sphere

Positive definiteness

A function $f : [-1,1] \to \mathbb{R}$ is positive definite on $\mathbb{S}^{n-1}$ if

$$\sum_{x,y \in C} f(\langle x, y \rangle) \geq 0$$

for every finite $C \subset \mathbb{S}^{n-1}$.

Gegenbauer basis

The cone has a natural basis: Gegenbauer polynomials

$$P_i^{(n)}(t), \qquad P_i^{(n)}(1) = 1,$$

orthogonal with weight

$$w(t) = (1 - t^2)^{(n-3)/2}.$$

Gegenbauer expansion

Every $f \in L^2([-1,1])$ has an expansion

$$f(t) = \sum_{i=0}^{\infty} f_i P_i^{(n)}(t),$$

with

$$f_i = \frac{\displaystyle\int_{-1}^{1} f(t) P_i^{(n)}(t) w(t)\, dt}{\displaystyle\int_{-1}^{1} \left(P_i^{(n)}(t)\right)^2 w(t)\, dt}.$$

Three-term recurrence

$$P_i^{(n)}(t) = \frac{(n+2i-4)tP_{i-1}^{(n)}(t) - (i-1)P_{i-2}^{(n)}(t)}{n+i-3}.$$

LP bound for T-avoiding codes

Theorem (Classical LP for codes)

Let $T \subset [-1, s)$ and let f be a polynomial such that

1. $f(t) \leq 0$ *for every* $t \in [-1, s] \setminus T$*;*
2. $f_0 > 0$ *and* $f_i \geq 0$ *for every* i *in the Gegenbauer expansion.*

Then every T-avoiding spherical s-code $C \subset \mathbb{S}^{n-1}$ satisfies

$$|C| \leq \frac{f(1)}{f_0}.$$

Idea of the proof

Sign of f on allowed inner products gives an upper estimate of $\sum f(\langle x, y \rangle)$ over $C \times C$; positive definiteness of f gives a lower estimate.

Moments and designs

For a code C, define the i-th moment

$$M_i(C) := \sum_{x,y\in C} P_i^{(n)}(\langle x, y\rangle).$$

Moment criterion

$$M_i(C) \geq 0, \qquad C \text{ is a } \tau\text{-design} \iff M_1(C) = \cdots = M_\tau(C) = 0.$$

Theorem (LP for T-avoiding designs)

If $f(t) \geq 0$ on $[-1,1] \setminus T$ and $f_i \leq 0$ for all $i \geq \tau + 1$, then every T-avoiding τ-design satisfies

$$|C| \geq \frac{f(1)}{f_0}.$$

LP for energy

Theorem (Energy lower bound)

Let h be a potential and let f be a polynomial satisfying

1. $f(t) \leq h(t)$ *for* $t \in [-1,1] \setminus T$*;*
2. $f_i \geq 0$ *for all* $i \geq 1$*.*

Then every T-avoiding code $C \subset \mathbb{S}^{n-1}$ with $|C| = N$ satisfies

$$\mathcal{E}^h(C) \geq N^2 \left(f_0 - \frac{f(1)}{N} \right).$$

Equality often prescribes the inner product set, distance distribution, and design properties of the extremal configuration.

How to find good polynomials?

The central problem is to construct a polynomial f satisfying the sign and Gegenbauer-coefficient constraints.

1. Reverse engineering

Start from a good code and force roots at its inner products.

2. Brute-force LP

Solve a discretized dual program and repair the candidate polynomial.

3. Levenshtein approach

Optimize over structured nonnegative factors using Lukács representations.

Method 1: reverse engineering

Start with a candidate optimal code C.

Interior and boundary inner products

Let $I_i \subset I(C)$ be the inner products in the interior of $[-1, s] \setminus T$, and let $I_b \subset I(C)$ be the boundary inner products.

First auxiliary polynomial

$$f(t) = \prod_{r \in I_i} (t - r)^2 \prod_{r \in I_b} (t - r).$$

- Double roots preserve the required sign.
- Boundary roots can appear with odd multiplicity.
- Multiplying by a small-degree factor often gives extra flexibility.

Reverse-engineered examples

A 48-dimensional design bound

A $(-1/3, -1/6) \cup (1/6, 1/3)$-avoiding 11-design in $\mathcal{S}^{47}$ satisfies

$$|C| \geq 52\,416\,000,$$

with equality from minimal vectors of an extremal even unimodular lattice in $\mathbb{R}^{48}$.

A 32-dimensional code bound

A $(0, 1/4)$-avoiding $(1/2)$-code in $\mathbb{S}^{31}$ that is at least a spherical 3-design satisfies

$$|C| \leq 146\,880.$$

Using extra moment vanishing

The vanishing of moments of C can relax the coefficient constraints and shrink the forbidden set T.

Example in $\mathbb{R}^{23}$

For an arbitrary

$$\left(-\frac{13}{23}, -\frac{7}{23}\right) \cup \left(\frac{5}{23}, \frac{6}{23}\right)\text{-avoiding } \frac{11}{23}\text{-code}$$

one obtains

$$|C| \leq 48\,600.$$

Equality is achieved by a code inside the Leech lattice.

Method 2: the dual LP

The dual program for the code bound is

$$
\begin{aligned}
\max c: \quad & \sum_{i=1}^{k} c_i P_i^{(n)}(1) \le 1 - c, \\
& \sum_{i=1}^{k} c_i P_i^{(n)}(t) \le -c, \qquad t \in [-1, s] \setminus T, \\
& c_i \ge 0, \qquad 1 \le i \le k.
\end{aligned}
$$

Resulting bound

$$|C| \le \frac{1}{c_{\max}}.$$

Discretization and repair

Replace the continuum condition by finitely many nodes:

$$\sum_{i=1}^{k} c_i P_i^{(n)}(t_j) \leq -c, \qquad j = 1, \ldots, K.$$

- Choose nodes t_j in $[-1, s] \setminus T$.
- Solve the finite LP.
- Check and repair inequalities on each interval $[t_j, t_{j+1}]$.

Role in practice

This is mainly a search and verification tool: it helps identify the right exact polynomial.

Brute-force example

The test program for dimension 32, $(0, 1/4)$-avoiding $(1/2)$-codes which are 3-designs is

$$\begin{aligned} \max c: \quad & \sum_{i=1}^{k} c_i P_i^{(n)}(1) \leq 1 - c, \\ & \sum_{i=1}^{k} c_i P_i^{(n)}(t) \leq -c, \quad t \in [-1, 0] \cup [1/4, 1/2], \\ & c_i \geq 0, \quad 4 \leq i \leq k. \end{aligned}$$

Numerical clue

With 500 nodes and degree 16,

$$1/c_{\max} = 146880.0125\ldots,$$

suggesting an exact bound of $146\,880$. On the other hand the milder condition of being 2-design (adding $c_3 \geq 0$) gives numerical bound 159835.084.

Brute-force example implemented

Theorem

Let $C \subset \mathbb{S}^{31}$ be a $(0, 1/4)$-avoiding $(1/2)$-code that is (at least) a spherical 3-design. Then $|C| \leq 146\,880$. This bound is achieved by minimal vectors of an extremal even unimodular lattice in $\mathbb{R}^{32}$.

There are many-many extremal lattices!

Method 3: Levenshtein approach for $T = (a, b)$

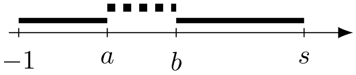
Consider

$$f(t) = (t-a)(t-b)(t-s)g(t),$$

where

$$g(t) \geq 0 \qquad \text{for } t \in [-1, 1].$$

- The cubic factor controls the sign on the allowed set.
- Nonnegativity of g is enforced structurally.
- Then one optimizes $f(1)/f_0$.



Lukács representation

Theorem (Lukács)

If $g(t) \geq 0$ *on* $[-1,1]$ *and* $\deg g = m$, *then*

$$g(t) = \begin{cases} U(t)^2 + (1-t^2)V(t)^2, & m \text{ even}, \\ (1+t)W(t)^2 + (1-t)Z(t)^2, & m \text{ odd}. \end{cases}$$

Easier (even) case

For $m = 2k$,

$$U(t) = \sum_{i=0}^{k} u_i P_i^{(n)}(t), \qquad V(t) = \sum_{i=0}^{k} v_i P_i^{(n)}(t).$$

Stationary equations

Optimize

$$\frac{f(1)}{f_0} = F(u_0, \ldots, u_k, v_0, \ldots, v_k)$$

over polynomials of degree at most $2k+3$.

Lagrange multiplier outcome

$$v_0 = \cdots = v_k = 0, \qquad R\mathbf{u} = \mathbf{j},$$

where $\mathbf{j} = (1, \ldots, 1)^T$ and

$$R_{ij} = \left\langle (t-a)(t-b)(t-s)P_i^{(n)}(t)P_j^{(n)}(t), 1 \right\rangle.$$

The three-term recurrence makes R a 7-diagonal matrix.

Extremal polynomial from the Levenshtein approach

The stationary polynomial has the form

$$f(t) = (t-a)(t-b)(t-s)\left(\sum_{i=0}^{k} u_i P_i^{(n)}(t)\right)^2.$$

Optimization property

It maximizes $f(1)/f_0$ among polynomials of degree at most $2k+3$ within this structured class.

Empirical phenomenon

For many good codes, the resulting extremal polynomial has nonnegative Gegenbauer coefficients and is therefore LP-admissible.

A 23-dimensional Levenshtein example

With $k = 2$, the method gives the following theorem.

Theorem (Example)

Let

$$T \in \left\{ \left(-\frac{3}{5}, -\frac{1}{3}\right), \left(-\frac{1}{3}, -\frac{1}{15}\right), \left(-\frac{1}{15}, -\frac{1}{5}\right) \right\}.$$

The cardinality of a T-avoiding $\frac{7}{15}$-code in $\mathbb{R}^{23}$ is at most

$$47\,104.$$

Sharpness

The bound is achieved by a code inside the Leech lattice.

From LP bounds to universal optimality

Once a good polynomial f is known, interpolate the potential h at the roots of f.

Hermite interpolation

Construct H_f so that

$$H_f(t) \leq h(t), \qquad t \in [-1,1] \setminus T.$$

Newton form

$$H_f(t) = h(t_1) + \sum_{r=1}^{\deg f} h[t_1, \ldots, t_{r+1}] \prod_{j=1}^{r} (t - t_j).$$

- Divided differences are nonnegative for absolutely monotone h.
- It remains to prove positive definiteness of the partial products.

Energy example in $\mathbb{R}^{32}$

Let h be absolutely monotone with $h^{(8)} > 0$ on $(-1, 1)$, and let

$$T = \left(-\frac{1}{2}, -\frac{1}{4}\right) \cup \left(\frac{1}{4}, \frac{1}{2}\right).$$

Theorem (Universal-optimality-type energy bound)

For a T-avoiding code $C \subset \mathbb{R}^{32}$,

$$E_h(C) \geq 146\,880 \Big[h(-1) + 1240 h\left(-\frac{1}{2}\right) + 31744 h\left(-\frac{1}{4}\right) + 80910 h(0) + 31744 h\left(\frac{1}{4}\right) + 1240 h\left(\frac{1}{2}\right) \Big].$$

Equality is attained by minimal-norm vectors of extremal even unimodular lattices in $\mathbb{R}^{32}$.

Direction 1: SDP methods

Semidefinite programming

SDP is a natural strengthening of LP for spherical codes.

- Bachoc–Vallentin developed an SDP method for spherical codes.
- The T-avoiding setting suggests analogous SDP formulations.
- The design case is especially open: SDP has not yet been systematically applied to spherical designs.

Question

Can SDP exploit the forbidden set T more efficiently than LP, especially for designs?

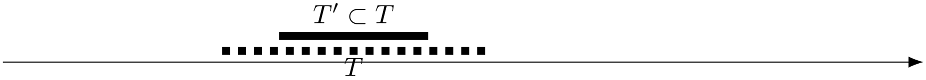
Direction 2: shrinking T

Suppose a sharp inequality is known for a forbidden set T.

Question

Find a minimal set $T' \subset T$ for which the same inequality remains valid.

- Can the interval endpoints be moved inward?
- Which forbidden intervals are genuinely necessary?
- Can equality cases force a minimal forbidden structure?



Direction 3: explicit Levenshtein-type formulas

For ordinary s-codes, Levenshtein obtained an explicit formula for the solution of the stationary system:

$$u_i = h_i \begin{vmatrix} P_i^{(n)}(s) & P_{k+1}^{(n)}(s) \\ P_i^{(n)}(1) & P_{k+1}^{(n)}(1) \end{vmatrix}, \qquad h_i = \dim \mathrm{Harm}_i(\mathbb{R}^n) = \binom{n+i-1}{i} - \binom{n+i-3}{i-2}.$$

Goal

Find an analogous explicit formula for (a,b)-avoiding s-codes. Prove the positive definiteness of an extremal polynomial.

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